PHASE 1 WIP — 2 sheets unimplemented. Main power-chain and safety gates OPEN.

01_USB_C_PD

文件: 01_USB_C_PD.kicad_sch

02_EXT_INPUT

文件: 02_EXT_INPUT.kicad_sch

03_EXTERNAL_POWER_MUX

文件: 03_EXTERNAL_POWER_MUX.kicad_sch

04_BATTERY_CHARGER

文件: 04_BATTERY_CHARGER.kicad_sch

05_BATTERY_PROTECTION

文件: 05_BATTERY_PROTECTION.kicad_sch

06_FUEL_GAUGE

文件: 06_FUEL_GAUGE.kicad_sch

07_UPS_POWER_MUX

文件: 07_UPS_POWER_MUX.kicad_sch

08_3V3_BUCKBOOST

文件: 08_3V3_BUCKBOOST.kicad_sch

09_5V_BUCKBOOST

文件: 09_5V_BUCKBOOST.kicad_sch

10_SENSOR_POWER

文件: 10_SENSOR_POWER.kicad_sch

11_MAINBOARD_INTERFACE

文件: 11_MAINBOARD_INTERFACE.kicad_sch

12_TEST_POINTS

文件: 12_TEST_POINTS.kicad_sch

13_MASTER_POWER_GATE

文件: 13_MASTER_POWER_GATE.kicad_sch

INCOMPLETE — NOT FOR FABRICATION

Sheet: /
File: FT02_PowerBoard_V1.kicad_sch

Title: FT02_PowerBoard_V1

Size: A4	Date:	**Rev: WIP-01**
KiCad E.D.A. 10.0.5		Id: 1/14

NOT IMPLEMENTED: HUSB238A cold-start 5V-only ordering/register proof: OI-008/009/020.

No PWR_FLAG, no ERC exclusions, no production footprint claims.

Intent is documented in docs/POWER_ARCHITECTURE.md and docs/OPEN_ISSUES.md.

This page has no electrical implementation. It must not be counted as a completed design.

INCOMPLETE — NOT FOR FABRICATION

Sheet: /01_USB_C_PD/
File: 01_USB_C_PD.kicad_sch

Title: 01_USB_C_PD

Size: A4	Date:	**Rev: WIP-01**
KiCad E.D.A. 10.0.5		Id: 2/14

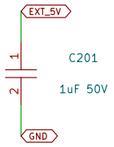
CANDIDATE WIRED: regulated EXT_5V ONLY; TPS259474L UV/OV/circuit breaker. Transient/source window OPEN.

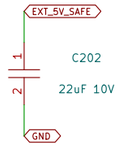
No PWR_FLAG, no ERC exclusions, no production footprint claims.



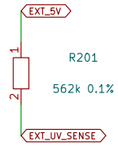
J201

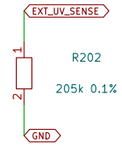
REGULATED EXT_5V

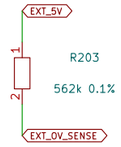


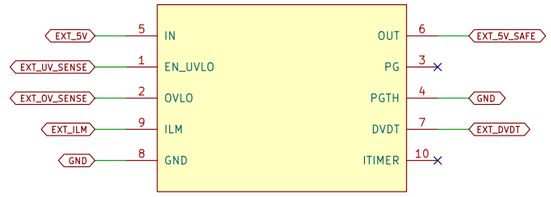






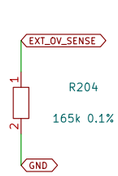


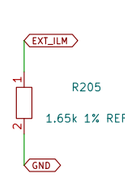




U201

TPS259474LRPWR CANDIDATE





IN=regulated external5V; OUT=EXT_5V_SAFE for future03 mux. No raw solar. Connector is logical, not released.

L variant latches overcurrent; UV recovery remains automatic. ITIMER NC selects fastest response, NOT a hard peak cap.

PG unused/NC,PGTH grounded. R201/R203 limit control-pin reverse current below10uA for a -5.5V test assumption.

Static OVP does NOT qualify5V+/-5%, hotplug overshoot or downstream5.5V safety. TVS/negative-clamp/thermal gates OPEN.

INCOMPLETE — NOT FOR FABRICATION

Sheet: /02_EXT_INPUT/
File: 02_EXT_INPUT.kicad_sch

Title: 02_EXT_INPUT

Size: A3	Date:	**Rev: WIP-01**
KiCad E.D.A. 10.0.5		Id: 3/14

NOT IMPLEMENTED: TPS2121 candidate; total input-current control and weak-USB fallback unresolved.

No PWR_FLAG, no ERC exclusions, no production footprint claims.

Intent is documented in docs/POWER_ARCHITECTURE.md and docs/OPEN_ISSUES.md.

This page has no electrical implementation. It must not be counted as a completed design.

INCOMPLETE — NOT FOR FABRICATION

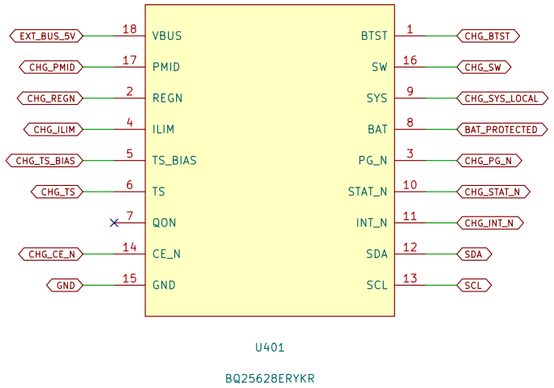
Sheet: /03_EXTERNAL_POWER_MUX/
File: 03_EXTERNAL_POWER_MUX.kicad_sch

Title: 03_EXTERNAL_POWER_MUX

Size: A4	Date:	**Rev: WIP-01**
KiCad E.D.A. 10.0.5		Id: 4/14

WIRED / CHARGE LOCKED OFF: local SYS only. R407 DNP and TS forced fault; source–priority controls OPEN.

No PWR_FLAG, no ERC exclusions, no production footprint claims.

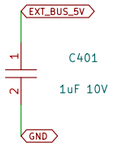


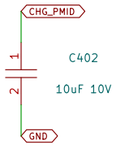
U401

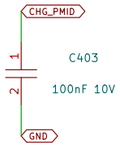
BQ25628ERYKR

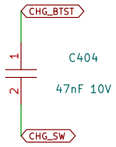
Q401

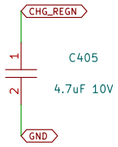
2N7002

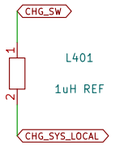


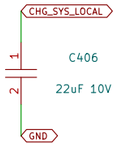


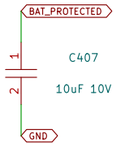


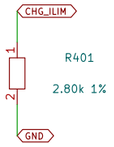


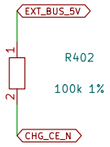


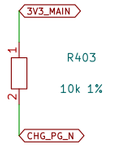


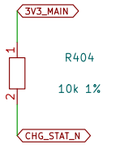


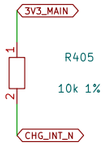


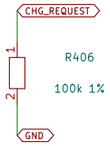


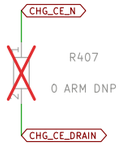


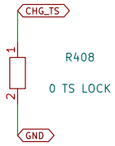


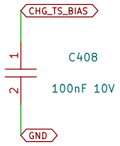












QON intentionally NC: internal pullup, no ship/reset button. SYS_LOCAL has NO FT02 system or debug loads.

CE pullup uses protected external input; R407 MUST remain DNP. TS=GND forces fault; this is NOT an NTC substitute.

R401 sets ~0.893A nominal branch input limit, NOT whole–board current. No charge profile/temperature approval.

Charger input leakage/reverse paths and hardware charge shedding remain review gates before enabling R407.

INCOMPLETE — NOT FOR FABRICATION

Sheet: /04_BATTERY_CHARGER/
File: 04_BATTERY_CHARGER.kicad_sch

Title: 04_BATTERY_CHARGER

Size: A3	Date:	**Rev: WIP–01**
KiCad E.D.A. 10.0.5		Id: 5/14

CANDIDATE WIRED: BQ29700 low-side guard. Startup, load/thermal and PCM coordination OPEN; charging locked OFF.

No PWR_FLAG, no ERC exclusions, no production footprint claims.



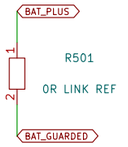
J501

1S PACK / LOGICAL



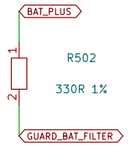
U501

BQ29700DSER CANDIDATE



R501

0R LINK REF



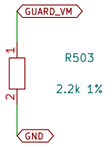
R502

330R 1%



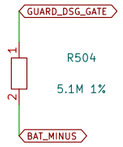
C501

100nF 10V



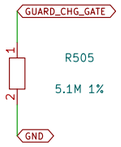
R503

2.2k 1%



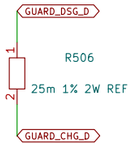
R504

5.1M 1%



R505

5.1M 1%



R506

25m 1% 2W REF



Q501

AON3402 CANDIDATE



Q502

AON3402 CANDIDATE

BAT_MINUS is isolated from system GND by Q501/R506/Q502. NEVER bridge it with a debug, shield or ground connection.

R501 is a positive power link; R601 is the ONLY gauge shunt. Protector bias/filter/gate drive bypass R601 and are audited.

R506 contributes to delayed overcurrent detection, NOT a5A clamp. First insertion may require charger/service arming.

Low-side opening does NOT isolate BAT_PLUS. Reverse insertion, current/thermal margins and fault leakage remain OPEN.

INCOMPLETE — NOT FOR FABRICATION

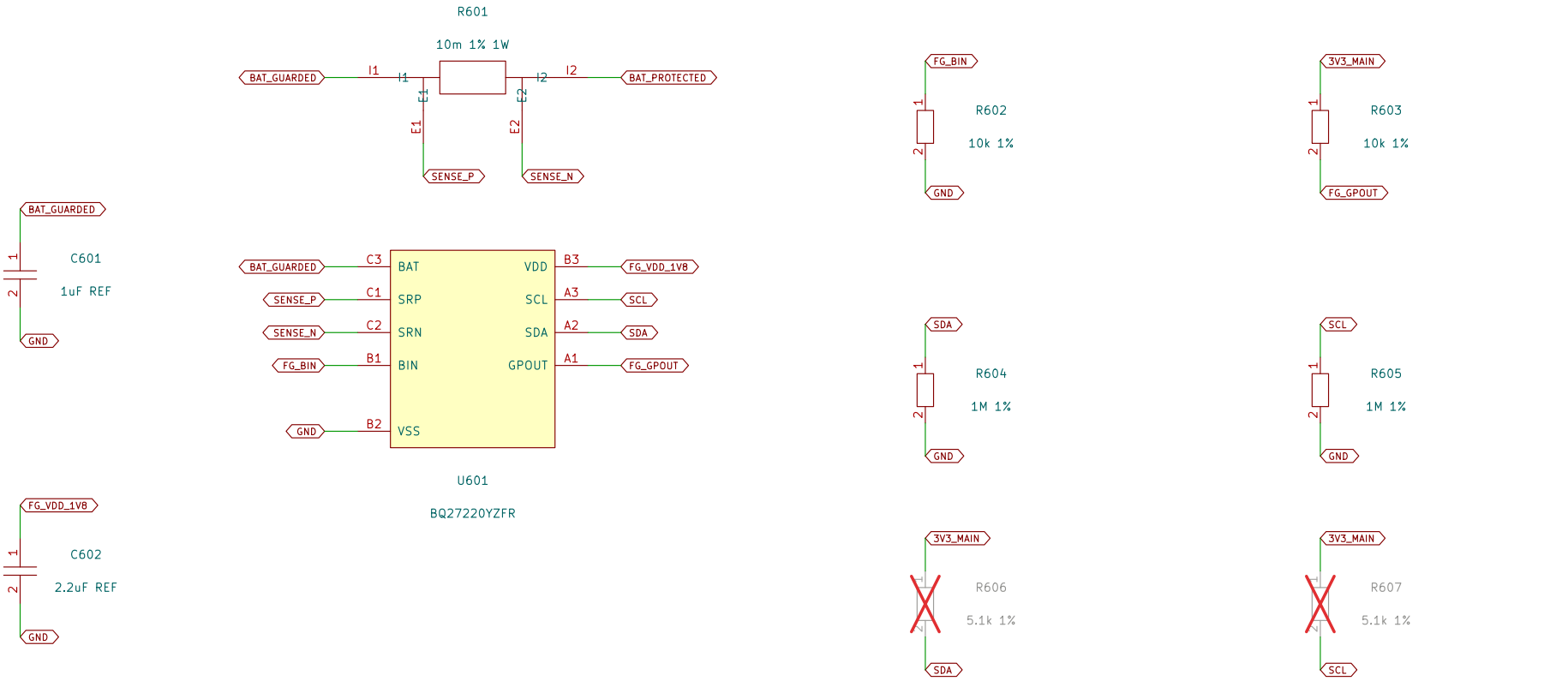
Sheet: /05_BATTERY_PROTECTION/
File: 05_BATTERY_PROTECTION.kicad_sch

Title: 05_BATTERY_PROTECTION

Size: A3	Date:	**Rev: WIP-01**
KiCad E.D.A. 10.0.5		Id: 6/14

REFERENCE ONLY: BQ27220 system-side high-side shunt. Auxiliary bypass/profile remain OPEN (OI-010).

No PWR_FLAG, no ERC exclusions, no production footprint claims.



I1/I2/E1/E2 are logical Kelvin terminals, NOT footprint pad numbers. Main charger/UPS attach to BAT_PROTECTED.

BAT_GUARDED: future protected-pack + board guard output. U601 BAT bias BYPASSES shunt; no full-current claim.

R602 is fixed BIN insertion bias, NOT an NTC. R606/R607 DNP; actual I2C pullups require MainBoard audit.

FG_GPOUT is private, NOT PWR_INT. VDD is decoupling only. Charge voltage/temperature remain OPEN.

5A pack ceiling: nominal shunt 50mV / 0.25W. No >5A pulse approval, no profile/thermal/hardware qualification.

INCOMPLETE — NOT FOR FABRICATION

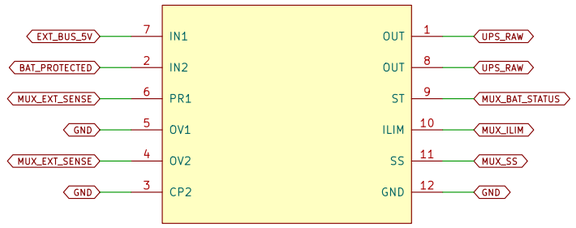
Sheet: /06_FUEL_GAUGE/
File: 06_FUEL_GAUGE.kicad_sch

Title: 06_FUEL_GAUGE

Size: A4	Date:	**Rev: WIP-01**
KiCad E.D.A. 10.0.5		Id: 7/14

CANDIDATE WIRED / UNFROZEN: TPS2121 and rail UVLO. Source qualification and UPS timing NOT proven.

No PWR_FLAG, no ERC exclusions, no production footprint claims.



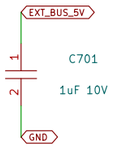
U701

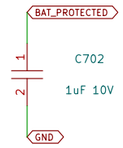
TPS2121RUXR UNFROZEN

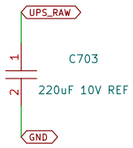


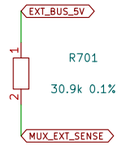
U702

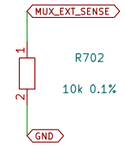
TPS3808G01DBVR

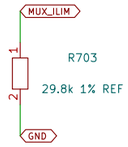


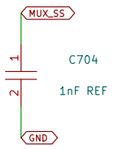


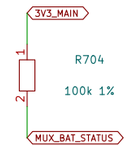


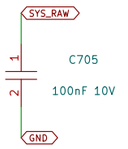


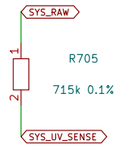


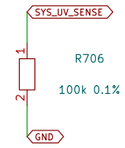


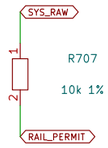












IN1=external,IN2=post-shunt battery. PR1/OV2 shared divider disables battery above external threshold (~4.34V).

CP2=GND:100us TYP transition, NOT5us and no guaranteed maximum. Voltage-only sense is NOT qualified EXT_VALID.

UVLO nominal SYS_RAW=3.30075V; CT NC=20ms typical recovery. This is NOT the pack cutoff or loaded-voltage spec.

ILIM reference=3/3.5/4A at29.8k (resistor tolerance additional). Load, thermal, rebound and hold-up gates OPEN.

INCOMPLETE — NOT FOR FABRICATION

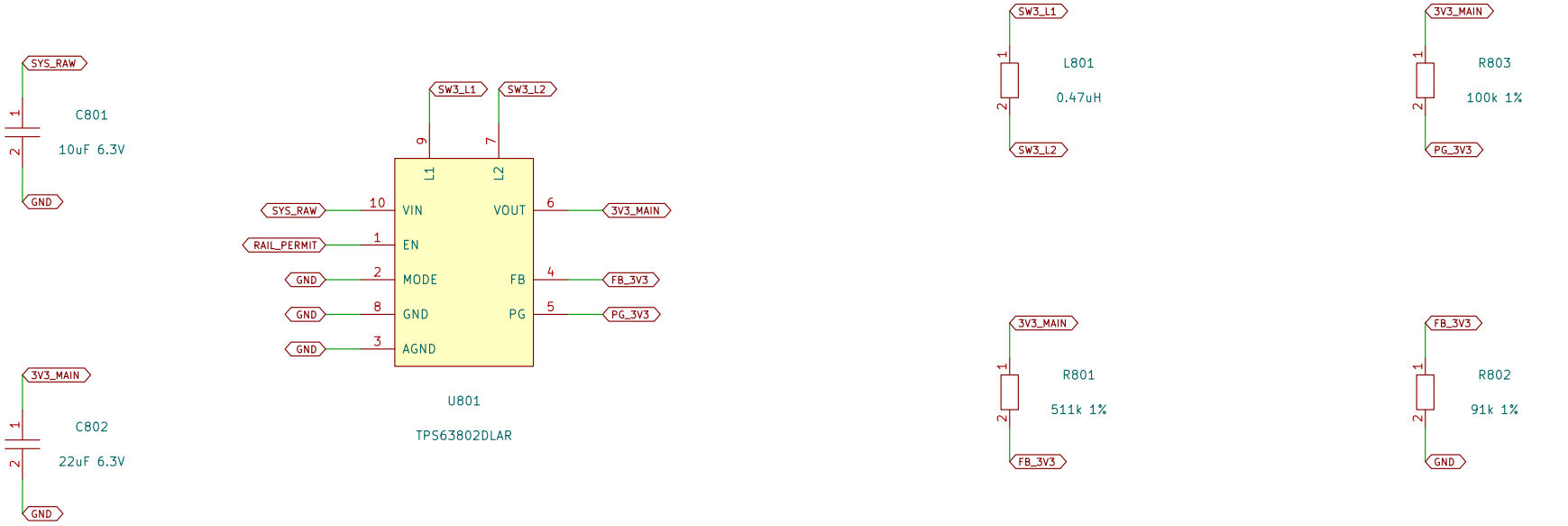
Sheet: /07_UPS_POWER_MUX/
File: 07_UPS_POWER_MUX.kicad_sch

Title: 07_UPS_POWER_MUX

Size: A3	Date:	Rev: WIP-01
KiCad E.D.A. 10.0.5		Id: 8/14

REFERENCE CIRCUIT ONLY: TPS63802 Fig10-1/Table10-6. NOT a qualified FT02 rail.

No PWR_FLAG, no ERC exclusions, no production footprint claims.



EN=RAIL_PERMIT from U702: automatic system UVLO, nominal3.30V. Battery cutoff and rebound NOT qualified.

PG pulled to 3V3_MAIN (not raw input) for Core voltage compatibility. Input protection remains mandatory.

Reference passives are not a released BOM. No transient simulation or full-temperature qualification performed.

INCOMPLETE — NOT FOR FABRICATION

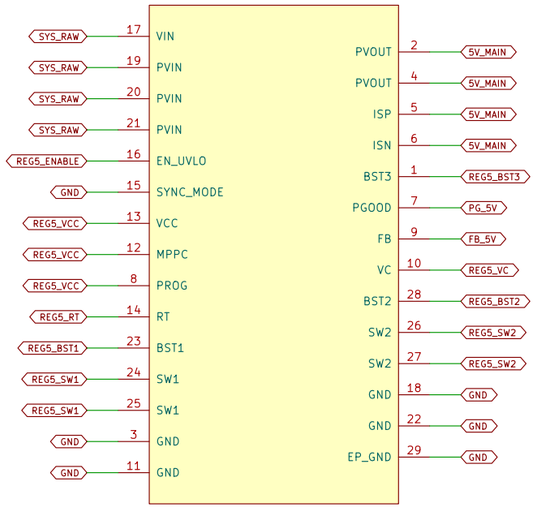
Sheet: /08_3V3_BUCKBOOST/
File: 08_3V3_BUCKBOOST.kicad_sch

Title: 08_3V3_BUCKBOOST

Size: A4	Date:	Rev: WIP-01
KiCad E.D.A. 10.0.5		Id: 9/14

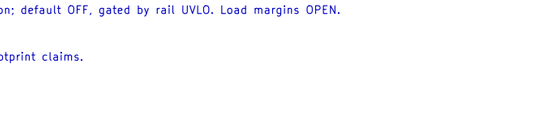
WIRED / UNQUALIFIED: LT3120 5V 500kHz application; default OFF, gated by rail UVLO. Load margins OPEN.

No PWR_FLAG, no ERC exclusions, no production footprint claims.



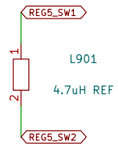
U901

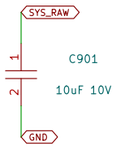
LT3120IUFDM

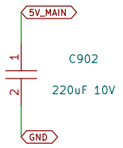


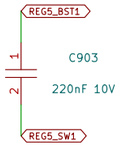
U902

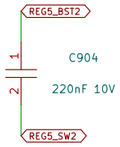
SN74LVC1G08DBVR

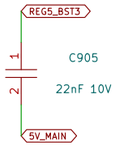


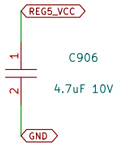


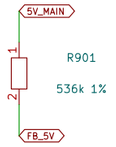


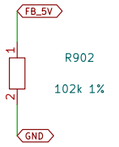


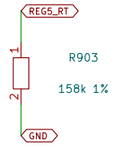


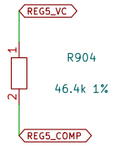


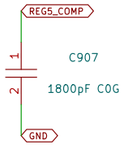


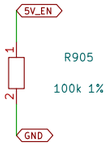


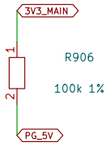


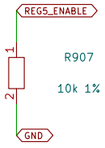


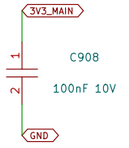












ADI page28 TA07 topology,500kHz/Burst. U902 ANDs Core5V_EN with RAIL_PERMIT; output pulldown holds OFF when unpowered.

ISP/ISN tied to PVOUT; PROG/MPPC tied to VCC. No programmed output current limit is claimed.

Reference example at VIN=2.5V supplies1.25A. This is NOT a qualified FT02 5V/2A or simultaneous rail rating.

U902 supply=3V3_MAIN; inputs tolerate5.5V. L/C selection, loop stability, UVLO timing and thermal qualification OPEN.

INCOMPLETE — NOT FOR FABRICATION

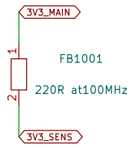
Sheet: /09_5V_BUCKBOOST/
File: 09_5V_BUCKBOOST.kicad_sch

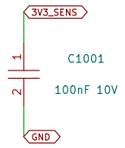
Title: 09_5V_BUCKBOOST

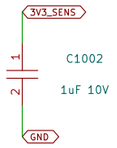
Size: A3	Date:	**Rev: WIP-01**
KiCad E.D.A. 10.0.5		Id: 10/14

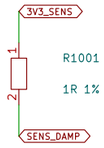
WIRED: 3V3_MAIN ferrite branch, decoupling and RC damping provision. EMI/load qualification OPEN.

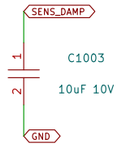
No PWR_FLAG, no ERC exclusions, no production footprint claims.

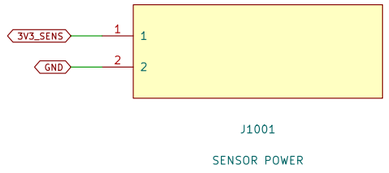












J1001

SENSOR POWER

Always-on filtered derivative of3V3_MAIN; no additional DC/DC. Optional load switch NOT fitted in this variant.

Ferrite source: Murata BLM21PG221SN1D,2A nominal rating,DCR<=45mohm; not a branch current limiter.

Series1R+10uF shunt branch is a damping provision. Actual bias-dependent impedance/load/EMI must be tested.

J1001 is a logical interface, not a released connector choice. SEN0695 module retains its own local1.8V regulator.

INCOMPLETE — NOT FOR FABRICATION

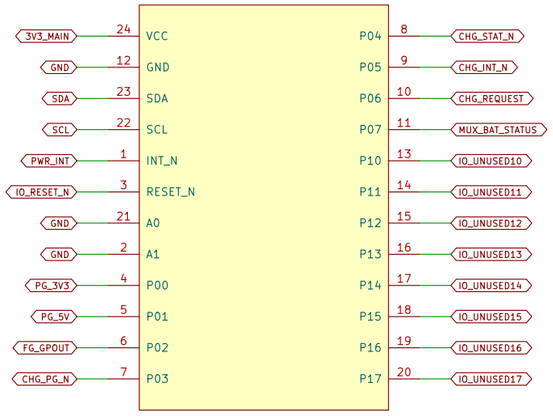
Sheet: /10_SENSOR_POWER/
File: 10_SENSOR_POWER.kicad_sch

Title: 10_SENSOR_POWER

Size: A3	Date:	**Rev: WIP-01**
KiCad E.D.A. 10.0.5		Id: 11/14

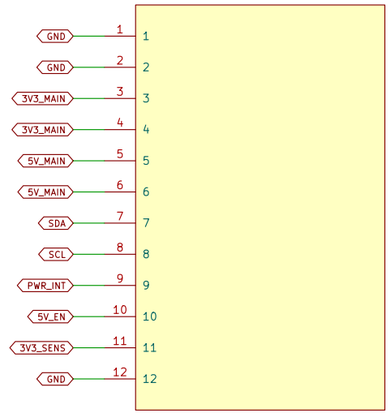
PARTIAL: logical connector and TCA9539 0x74 status I/O. No new MCU. Source-policy integration OPEN.

No PWR_FLAG, no ERC exclusions, no production footprint claims.



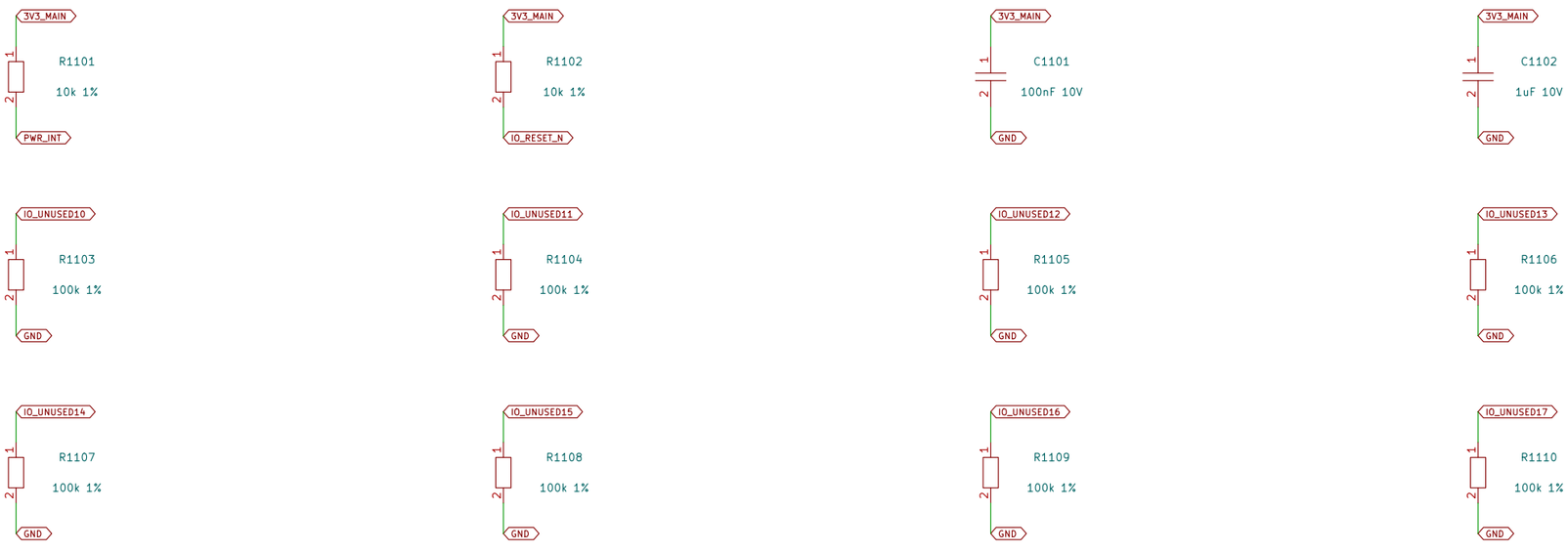
U1101

TCA9539PWR



J1101

MAINBOARD LOGICAL



A1=A0=0:0x74. At POR all ports input. CHG_REQUEST external pulldown and R407 DNP prevent unintended charge.

P00/P01=rail PG;P02=gauge;P03..P05=charger status;P06=charge request;P07=MUX ST. P10..P17 have100k pulldowns.

PWR_INT is expander interrupt only; gauge wake needs P02 drive-low then input release. NEVER drive-high.

Connector/pullups/off-state behavior and source-policy I/O remain OPEN. No GPIO4 allocation; Core uses SDA47/SCL21.

INCOMPLETE — NOT FOR FABRICATION

Sheet: /11_MAINBOARD_INTERFACE/
File: 11_MAINBOARD_INTERFACE.kicad_sch

Title: 11_MAINBOARD_INTERFACE

Size: A3	Date:	**Rev: WIP-01**
KiCad E.D.A. 10.0.5		Id: 12/14

LOGICAL TEST-POINT INVENTORY: most nets have no implemented source. Footprints NOT assigned.

No PWR_FLAG, no ERC exclusions, no production footprint claims.



INCOMPLETE — NOT FOR FABRICATION

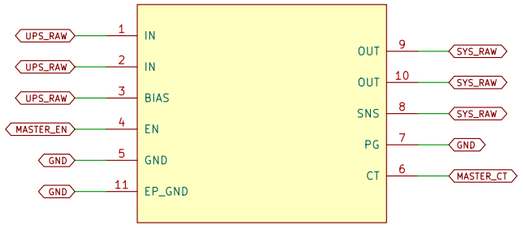
Sheet: /12_TEST_POINTS/
File: 12_TEST_POINTS.kicad_sch

Title: 12_TEST_POINTS

Size: A4	Date:	**Rev: WIP-01**
KiCad E.D.A. 10.0.5		Id: 13/14

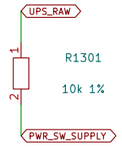
CANDIDATE WIRED: mechanical SYSTEM HARD OFF, upstream of every system rail. Backfeed/leakage qualification OPEN.

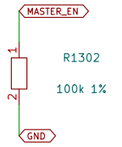
No PWR_FLAG, no ERC exclusions, no production footprint claims.

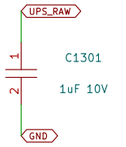


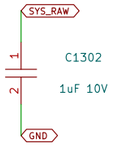
U1301

TPS22953DQCR CANDIDATE

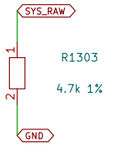


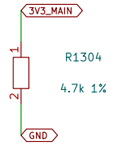


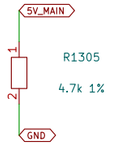


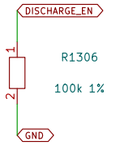


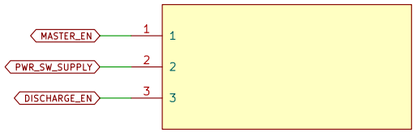






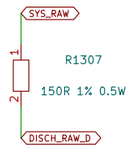


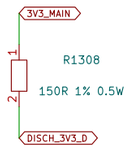


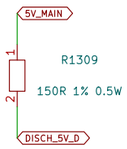


J1301

PWR_SW: ON / COMMON / OFF









Q1301

DMN2056U-7



Q1302

DMN2056U-7



Q1303

DMN2056U-7

SPDT COMMON2 to ON1: MASTER_EN high. To OFF3: DISCHARGE_EN high; MASTER_EN pulled low. Open=passive decay.

IN+BIAS=UPS_RAW; OUT=SYS_RAW. No MCU/I2C control. Charger and gauge remain upstream and powered.

Three separate150R+NMOS discharge branches need upstream bias;4.7k bleeders remain if all sources vanish.

Overlap is resistor-limited, NOT assumed absent. Contact/ESD, hot leakage, resistor pulse and rail injection gates OPEN.

INCOMPLETE — NOT FOR FABRICATION

Sheet: /13_MASTER_POWER_GATE/
File: 13_MASTER_POWER_GATE.kicad_sch

Title: 13_MASTER_POWER_GATE

Size: A3	Date:	**Rev: WIP-01**
KiCad E.D.A. 10.0.5		Id: 14/14